Hybrid Integration of Gate-All-Around Stacked Si Nanosheet FET and Si/SiGe Super-Lattice FinFET to Optimize 6T-SRAM for N3 Node and Beyond

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***Abstract*— The implementation of vertically stacked gate-all-around nanosheet FET (GAA NSFET) may help improve the performance of static random access memory (SRAM) for the design flexibility with variable NS widths. However, the method that often relies on the increasing of device width for higher driving current is to cause an increased SRAM cell area and degrade the scalability of transistor in advanced nodes. In this article, an innovated SRAM bitcell design with hybrid integration of Si NSFET and Si/SiGe super-lattice FinFET (SL-FinFET) in one SRAM cell is proposed for less area and improved performance in 3 nm and beyond nodes. Under the same footprint, p-type SL-FinFET delivers obviously enhanced driving current compared with NSFET and it may yield improved circuit performance for the SRAM cell with natural transistor ratio. The integration process method and the layout schemes for two kinds of typical 6T-SRAM cells, including the high-robustness cell (HI-R Cell) and the high-speed cell (HI-S Cell) with the hybrid integration method, are presented for optimizing high-density (HD) and high-performance (HP) SRAM. As compared with the HD NSFET SRAM cell, the HD HI-R Cell achieves 16.2% improvement in read static noise margin (RSNM) and the HD HI-S Cell achieves 21.6% reduction in read time and 28.6% reduction in write time. The HP hybrid integrated cell even pushes the SRAM cell performance arriving to the theoretical limits of regular NSFETs. Moreover, the hybrid integration scheme may provide another revolutionary method to optimize transistor ratio by adjusting the height of SiGe thickness in the transistors.**

***Index Terms*— 3-nm process node, 6T static random access memory (6T-SRAM), hybrid integration, Si/SiGe super-lattice FinFET (SL-FinFET), vertically stacked gate-all-around nanosheet FET (GAA NSFET).**

I. Introduction

LARGE volume and fast static random access memory (SRAM) help the computing system boost performance and occupy a considerable portion of the die size in many computing chips. Therefore, the development of higher density SRAM with less bitcell area as well as stronger driving ability inside is one of the most critical tasks to maintain the scaling of CMOS technology in past decades [1]. The bitcell area of standard 6T-SRAM has already shrunk to 0.021 μm^2 at 5 nm node for updated FinFET process [2].However for 3 nm or beyond nodes, the gate-all-around nanosheet FETs (GAA NSFETs) have proven to be a promising successor of the FinFET, with better electrostatic in channel and stronger device driving performance due to superior gate control capability on vertically stacked nanosheet channels [3], [4], [5], [6], [7], [8]. In addition, GAA NSFET offers the design flexibility that was sacrificed partly while FinFETs replace traditional planar transistors [9]. For high-density (HD) SRAM bitcells, FinFET must adopt 1:1:1 or 1:1:2 fin configurations with the quantized width design limitations, which often results in the degradation of bitcell performance, such as the conflict of read and write stability [10]. The novel NSFET can obtain 1: α: β configuration to achieve improved bitcell specifications, such as the increasing of the disturb margin [11].

A series of device, process, and design optimization technologies for new NSFETs building HD and high-performance

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(HP) SRAM bitcells and circuits are reported [12], [13], [14]. Huynh-Bao et al. [15], [16] proposed device threshold voltage retargeting technology to reduce the minimum operating voltage (V_{min}) of SRAMs with NSFETs. Vashishtha [17] have done a lot of research on the optimization of GAA NSFET parameters on SRAM performance. However, it is also found that a larger parasitic capacitance in GAA NSFETs introduced by its more complex 3-D channel structure makes a longer read and write delay in SRAM bitcell [18], [19], [20], [21], [22], [23]. Especially for HD SRAM with 1:1:1 configuration with extremely scaled nodes, the flexible increase of nanosheets width may improve read and write margin or driving ability but obviously increase the cell area, which perhaps awakes a great challenge for the NSFET competing with the cutting-edge FinFET technology [24].

In order to obtain the best SRAM bitcell performance and the most aggressively scaling area with GAA NSFETs at the same time, an innovative approach by hybrid integrating Si NSFET with Si/SiGe super-lattice FET (SL-FinFET) in 6T-SRAM cell is firstly proposed in the article. The SL-FinFET was formed by no nanosheets release process implemented in replacement metal-gate (RMG) [25] and applied into the hybrid integration of high-voltage I/O devices with NSFET core devices [26] to overcome the challenge of thick gate insulator filling between stacked Si NSs. The stronger driving ability [27], [28], [29], [30] than NSFET will reduce the area cost of obtaining greater driving strength and cope with the high drive requirements.

In the article, at the first time, the hybrid integration scheme for combining Si NSFET with Si/SiGe SL-FinFET in one 6T-SRAM bitcell (HI-SRAM) is proposed, which may effectively take both advantages of these two devices to optimize the critical circuit performance at scaled process nodes. The integration flow totally compatible with mainstream NSFETs manufacturing process is regulated and the electrical characteristics of new SRAM bitcell at 3-nm node are carefully evaluated by technical computer-aided design (TCAD) and custom circuit emulation tool (SPICE). With the revolutionary structure and optimized parameters, the speed and stability have been obviously improved. The results indicated that it is an effective approach to further boost the gain of power, performance, and area (PPA) in future circuits by GAA NSFET processes.

II. Hybrid Integration Method for SRAM

A. Hybrid Integration Scheme for SRAM

In general custom-integrated circuit (IC) design, the regulations of channel widths in transistors are critical for circuit performance optimizations. For the 6T-SRAM bitcell design, it often requires a ratio of pull-down (PD) transistor width (W_{PD}) > access transistor width (W_{AX}) > pull-up (PU) transistor width (W_{PU}) for better performance [31]. With the guide of the design method, in Si NSFET SRAM cell as shown in Fig. 1(a), the value of W_{PD}: W_{AX}: W_{PU} is flexibly adjusted and fixed at 1: α: β configuration, which may cause a slight increasing of cell area as compared with the natural design with W_{PD}: W_{AX}: W_{PU} = 1:1:1. In the general NSFET

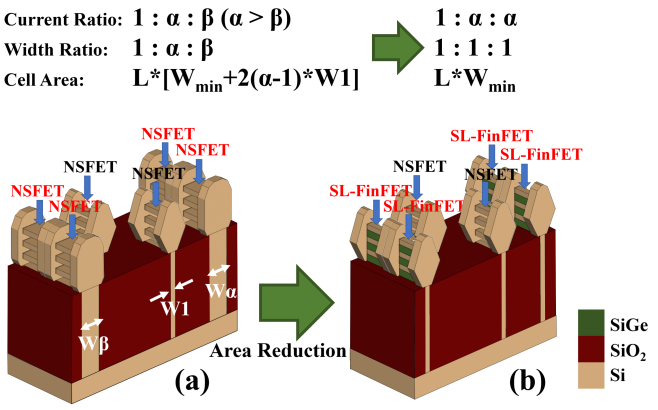


Fig. 1. Bitcell scheme for (a) NSFET SRAM and (b) hybrid integrated SRAM.

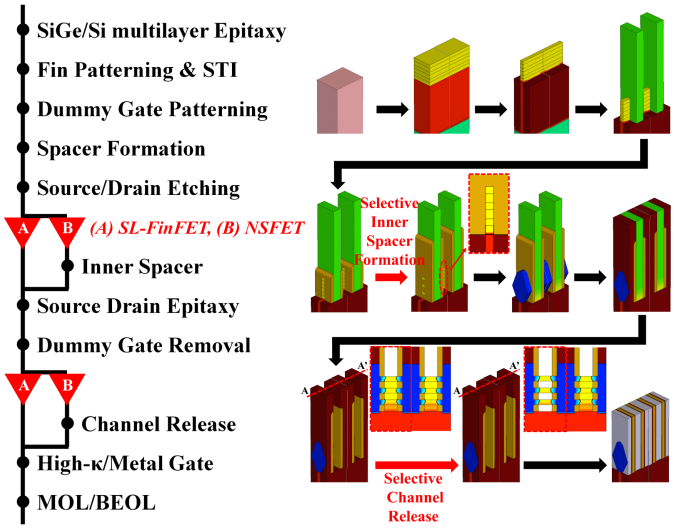


Fig. 2. Designed process flow of hybrid integration. Different from traditional NSFET, mainstream processes are marked with red arrows.

manufacturing process, the SL-FinFETs are easily formed by avoiding the nanosheets release process with protective mask and provide higher driving current. With different transistor driving ability between the NSFET and the SL-FinFET, the proposed hybrid integration of these two kinds of devices in SRAM cell is expected to provide the desired performance by following the guide of $W_{PD} > W_{AX} > W_{PU}$ with the natural design of W_{PD}: W_{AX}: W_{PU} = 1:1:1 [see Fig. 1(b)] for obtaining the minimum cell area in high-density SRAM circuits.

B. Process of Hybrid Integration

The designed process flow of hybrid integration of the GAA NSFET and the SL-FinFET on one Si wafer is shown in Fig. 2. At first, the epitaxy of the stacked SiGe/Si layers is grown on the bulk Si substrate, where the thickness of the Si layer is defined as the single-layer nanosheet channel thickness of the NSFET, and the total thickness of the SiGe/Si layers is the Fin channel height of the SL-FinFET. Then, fins, shallow trench isolation (STI), dummy gate, and spacer are formed in sequence with general GAA NSFET processes [32]. In the next steps, the source/drain (S/D) is etched by anisotropic methods

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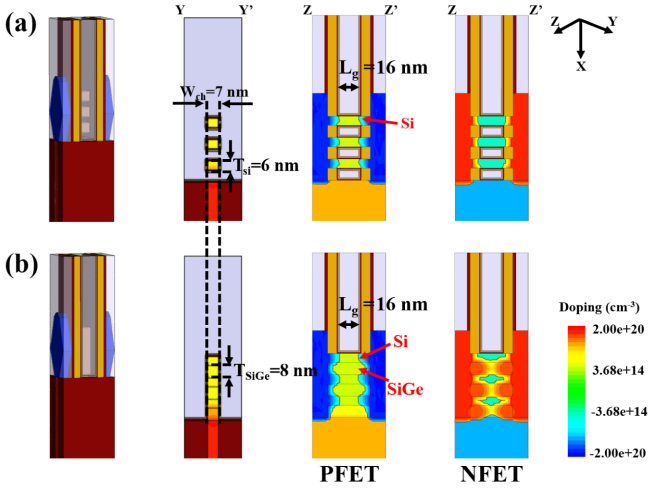


Fig. 3. Structure of the simulation domain of the 3-nm node (a) NSFETs and (b) SL-FinFETs.

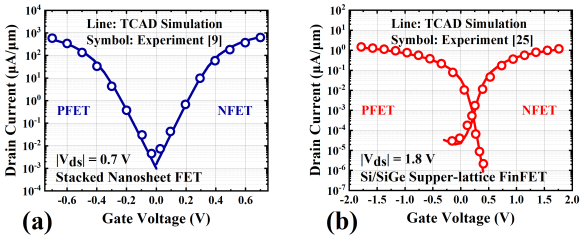


Fig. 4. I_D–V_G calibration of (a) stacked Si NSFET and (b) Si/SiGe super-lattice FinFET.

to the Si substrate and a shadowing mask is used to protect the SiGe channels in the SL-FinFET region, while the SiGe in the NSFET area is etched back and the conventional inner spacer is formed. After the removal of the shadowing mask, the embedded selective SiGe or SiC epitaxy with p-type and n-type in-suit doping is applied to form the S/D regions of FETs. Then, the zero-level isolation layer (ILD-0) is deposited and planarized were performed to expose the dummy gate, followed by the selective removal of the dummy gate stack (poly-Si/gate oxide). For the second time, the shadowing mask is formed in the SL-FinFET region to protect the SL-Fin structure again. Next, the selective etching of SiGe in the NSFET region is performed to release the Si nanosheet channel. Finally, the shadowing mask is removed and the RMG with multilayer HKMG process as well as the transistor contacts and the back-end-of-line (BEOL) process followed the conventional process used for traditional NSFETs, are adopted for the formation of devices and interconnections in SRAM circuits.

This designed process is easy to realize a hybrid integration of SL-FinFET on conventional GAA NSFET flow only with twice additional shadowing mask processes, which is the modification of the conventional NSFET manufacture process only to a small extent and a little extra production cost. Moreover, the effective width of SL-FinFET can be adjusted by changing the SiGe thickness in the Si/SiGe epitaxy process, without altering the effective width of NSFET, which provide another method to adjust the effective transistor width without the increase of designed channel width for regulation of the ratio between W_{PD}, W_{AX}, and W_{PU}.

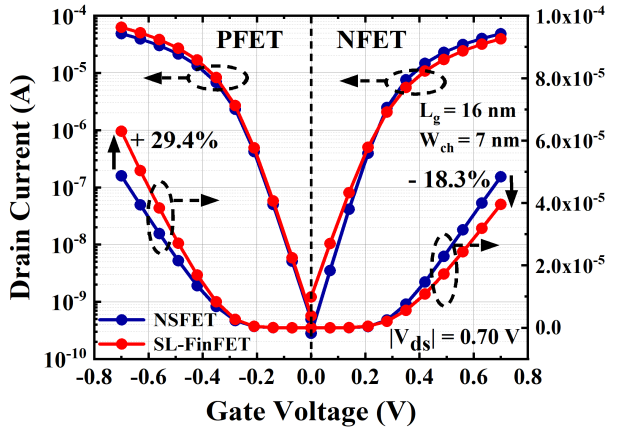
III. Device Structure and Performance

Fig. 3 shows the schematic of 3-nm-node NSFETs and SL-FinFETs formed by the designed process flow. PFET adopts SiGe S/D and NFET adopts SiC S/D. The main geometrical and process parameters for the NSFET are referred to the specification defined in International Roadmap for Devices and Systems (IRDS) 2022 [33]. Both NSFETs and SL-FinFETs are with the gate length (L_g) of 16 nm, channel projection width (W_{ch}) of 7 nm, and effective gate oxide thickness (EOT) of 0.808 nm. The thickness of the Si channel (T_{Si}) is 6 nm, and the thickness of the SiGe channel (T_{SiGe}) is 8 nm. There is a three-layer vertically stacked Si NS channel in one NSFET. Therefore, the total fin height of SL-Fin is 42 nm. The Ge concentration of SiGe in SL-Fin channels is set to 50% to significantly improve the mobility of the carriers. Besides, the NSFETs achieve (100) channel surface orientation and obtain higher electron mobility, while the hybrid integrated SL-FinFETs on the same wafer achieve (110) channel surface orientation and obtain higher hole mobility.

The electrical characteristics of both 3-nm-node Si NSFET and Si/SiGe SL-FinFET devices are simulated by Synopsys Sentaurus TCAD. The process simulation has considered the Ge mole fraction and the Ge intermixing caused by thermal annealing. The common drift-diffusion model solved by the Poisson and carrier continuity equation is adopted. The quantum effect, bandgap narrowing, band-to-band tunneling, carrier generation, and recombination are considered. To simulate the mobility degradation, the remote phonon and coulomb scattering at the channel and insulator interface are calculated, and the inversion and accumulation layer mobility models are used to calculate impurity, phonon, and surface roughness scatterings. At the same time, the velocity saturation model is implemented for high gate and drain bias. In addition, the Piezo model is introduced due to the lattice interaction between Si and SiGe [34]. Advanced calibrations of Si and SiGe material parameters [27], [35], [36], [37] have been performed for the unique super-lattice channel of the SL-FinFET. The TCAD simulation results for both NSFETs and SL-FinFETs are well calibrated with the experimental data of stacked nanosheet GAA transistors [9], [33] and Si/SiGe super-lattice FinFETs [25], as shown in Fig. 4, which ensures the simulation accuracy of the devices.

The operating voltage of simulated devices is fixed at 0.7 V. The thresholds of NFET and PFET are tuned to symmetrical 0.22 and −0.22 V, respectively. The simulation results showed a significant performance improvement in p-type SL-FinFET compared with p-type NSFET, particularly in driving ability, as shown in Fig. 5, and obtaining nearly 30% higher than that of the p-type NSFET. The significant driving current improvement is attributed to the enhanced hole mobility due to the (110) channel surface orientation of the SL-FinFET, the remained SiGe channel between Si NSs, and the compressive strain introduced by the superlattice stack structure in SL-FinFET. However, the short-channel effect of p-type SL-FinFET has slightly increased, resulting in degradation in subthreshold swing (SS),

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	p NSFET	p SL-FinFET	n NSFET	n SL-FinFET
SS (mV/dec)	69.2	70.0	64.1	75.4
DIBL (mV/V)	30.8	50.8	23.1	73.8

Fig. 5. Transfer characteristics of the 3-nm node NSFETs and SL-FinFETs at 0.7 V operating voltage.

drain-induced barrier lowering (DIBL), and OFF-current (Ioff). For NFET, transitioning from the GAA to the triple gate weakens the gate control ability, and the rapid diffusion of S/D doping in the channel significantly exacerbates the short-channel effect of the n-type SL-FinFET. So, it can be found that the performance of n-type SL-FinFET in all aspects such as ON-current (Ion), OFF-current (Ioff), SS, and DIBL is degraded. Therefore, the optimization to the SRAM cell with hybrid integration mainly focuses on the combined use of NSFET and p-type SL-FinFET in the SRAM circuit design.

IV. Hybrid Integrated SRAM Cell Design

In general, the 6T-SRAM cell consists of two PD NFETs and two PU PFETs to form two interlocked CMOS inverters for bit information storage, where two access (AX) transistors controlled by the word line (WL) and connected to complementary bit lines (BLs) are also included for bit information read/write control. In standard SRAM with GAA NSFET technology, NFETs are typically employed as AXs due to their higher driving current compared to PFETs under the same footprint. The n-type AX SRAM cell exhibits greater read stability with a larger PD ratio (the driving ability ratio of PD to AX) and higher write stability with a smaller PU ratio (the driving ability ratio of PU to AX). In the proposed SRAMs with hybrid integration, directly replacing p-type NSFETs with high-performance p-type SL-FinFETs as PU FETs may cause abnormal writing functions due to the increased PU ratio. Therefore, the design of p-type AX, which was also reported in earlier SRAM technology [38], was adopted instead. The incorporation of Si/SiGe SL-FinFET technology helps overcome the performance limitations of PFETs. In contrast to the n-type AX SRAM cell, the p-type AX SRAM cell may obtain higher read stability with a larger PU ratio and greater write stability with a smaller PD ratio.

Various hybrid integrated SRAM cells have been developed for different applications. One of the primary functions of SRAM is data exchange with CPU registers, and its speed directly impacts the maximum computing speed of the CPU. The driving strength of the AX plays a crucial role in determining the speed of SRAM. In light of this, a high-speed cell (HI-S Cell) has been designed, utilizing the hybrid integration method and employing p-type AXs, as shown in Fig. 6(b). To achieve enhanced speed, the HI-S Cell utilizes p-type SL-FinFET with high driving current as the AX. Additionally, to ensure proper read operations, p-type SL-FinFET is also employed as the PU, maintaining an optimal PU ratio. Furthermore, n-type NSFET is utilized as the PD to minimize leakage.

Moreover, certain SRAMs may encounter complex usage scenarios with high levels of external noise, demanding strong anti-interference capabilities. To address this requirement, a high-robustness cell (HI-R Cell) has been designed employing p-type AXs, as shown in Fig. 6(c). In this design, the AX utilizes p-type NSFET, while the PU employs p-type SL-FinFET. By leveraging the larger driving capability of the p-type SL-FinFET compared to the p-type NSFET, the PU ratio has been significantly enhanced, resulting in improved read stability.

The technology assumptions for the 3-nm-node SRAM are contacted gate pitch (CGP) = 45 nm, metal pitch (MP) = 30 nm, and channel spacer (CS) = 38 nm. The minimum channel width for both the NSFET and SL-FinFET is 7 nm, and the width of the nanosheets can vary within the range of 7–29 nm. SRAM cells are typically classified into HD cells and HP cells, depending on their intended usage. To achieve the most compact transistor arrangement and occupy the smallest cell area, HD SRAM cells employ the narrowest channel width. All three types of HD SRAM cells are compatible with the same manufacturing process, resulting in a consistent HD cell area of 90 × 180 nm^2, as shown in Fig. 6. Conversely, HP SRAM cells are designed to prioritize performance, even at the expense of increased area. In the HP NSFET SRAM design, the NSFET can be configured in 1: α: β setup, allowing for more flexible performance regulation. In the HP HI-SRAM design, the SL-FinFET follows a quantized width design, resulting in a fixed HP HI-SRAM cell area of 90 × 224 nm^2. The HP HI-S Cell aims to maximize the AX width to further enhance speed, while ensuring that the PU width remains consistent with the AX width to maintain normal read operation. The HP HI-R Cell is designed to maximize the PU ratio to further improve read stability.

V. Results and Discussions

The comparison and evaluation of the transient and static characteristics of the SRAM cells were performed through SPICE simulation. Compact models were created by fitting the physical parameters in the Berkeley short-channel insulated-gate field effect transistor model–common multigate (BSIM-CMG). Hold static noise margin (HSNM) and read static noise margin (RSNM) were extracted from the maximum inscribed square of each butterfly curve in the hold and read states. Additionally, write static noise margin (WSNM) was determined as the width of the minimum square that could be embedded between the lower-right half of the curves in the write states [39]. Read time (RT) is defined as the delay

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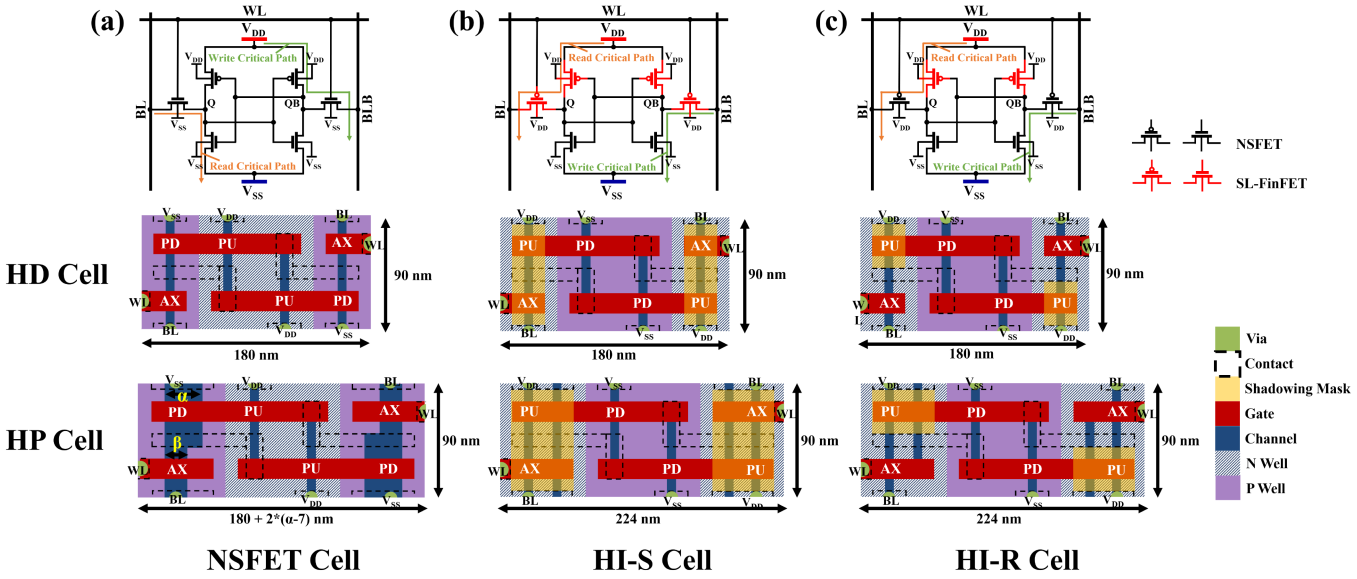


Fig. 6. Circuit structure and layout design of (a) NSFET SRAM cell, (b) hybrid integrated high-speed SRAM cell, and (c) hybrid integrated robust SRAM cell. The HD cell area for all types of SRAM cells is 90 × 180 nm.

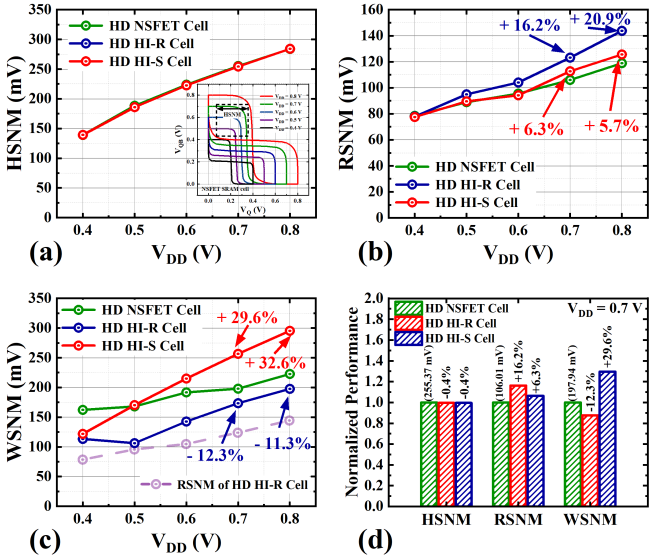


Fig. 7. Variation of (a) HSNM, (b) RSNM, and (c) WSNM of HD 6T-SRAM cells with V_{DD}. (d) Normalized performance of HD 6T-SRAM cells at 0.7 V operating voltage. The HI-R Cell balances the stability of reading and writing well.

between 50% WL signal rising/dropping edge (corresponding to the n-type and p-type AX SRAM cells, respectively) and 90% V_{DD} drop/rise of BLs. Similarly, write time (WT) is defined as the delay between 50% WL signal rising/dropping edge and 90% V_{DD} drop/rise of memory node.

A. HD SRAM Optimization

Fig. 7 shows the HSNM, RSNM and WSNM of the HD SRAM Cells. When the SRAM cell is in holding state, the AX is cutoff and the bi-stable circuit composed of two inverters is insensitive to the transistor ratio. As a result, the HSNM of the three SRAM cells remains largely consistent [see Fig. 7(a)]. The RSNM of the three HD SRAM cells exhibits a notable disparity, as shown in Fig. 7(b). In comparison to the HD NSFET Cell, the HD HI-R Cell demonstrates a noteworthy 16.2% improvement in RSNM at the operating voltage of 0.7 V. This improvement is attributed to the utilization of different types of devices for the PU and AX in the HD HI-R Cell. Consequently, the PU ratio increases to 63:48, surpassing the PD ratio of the HD NSFET Cell (48:48), resulting in improved RSNM. However, it should be noted that the SL-FinFET exhibits weaker gate control capability and more pronounced DIBL effect compared to the NSFET. As a consequence, the driving ability improvement provided by the SL-FinFET diminishes gradually at lower operating voltages, ultimately causing the disappearance of the RSNM improvement in the HD HI-R Cell under such conditions. The WSNM exhibits an inverse trend compared to RSNM, as shown in Fig. 7(c). In the HD HI-R Cell, there is a slight reduction in WSNM compared to the HD NSFET Cell. This trade-off involves sacrificing WSNM to enhance RSNM, aligning both parameters to reach a closer level, which balances the conflict between read and write stability and leads to a considerable improvement in the overall disturb margin. In contrast, the HD HI-S Cell benefits from a larger AX width, resulting in a substantial enhancement of WSNM. At the operating voltage of 0.7 V, the HD HI-S Cell achieves a significant 29.6% improvement in WSNM compared to the HD NSFET Cell.

Fig. 8 shows the RT and WT characteristics of the HD SRAM Cells. At the operating voltage of 0.7 V, the HD HI-S Cell demonstrates a 21.6% reduction in RT and a 28.6% reduction in WT compared to the HD NSFET Cell. This notable improvement in speed is primarily attributed to the implementation of p-type SL-FinFET, which accelerates the charging and discharging process of storage nodes and BLs

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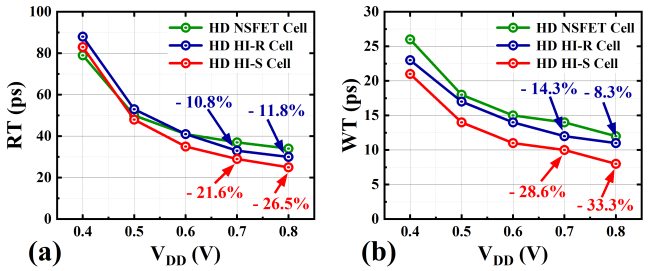


Fig. 8. Variation of (a) RT and (b) WT of HD 6T-SRAM cells with V_{DD}. The HI-S Cell shows great improvement in speed.

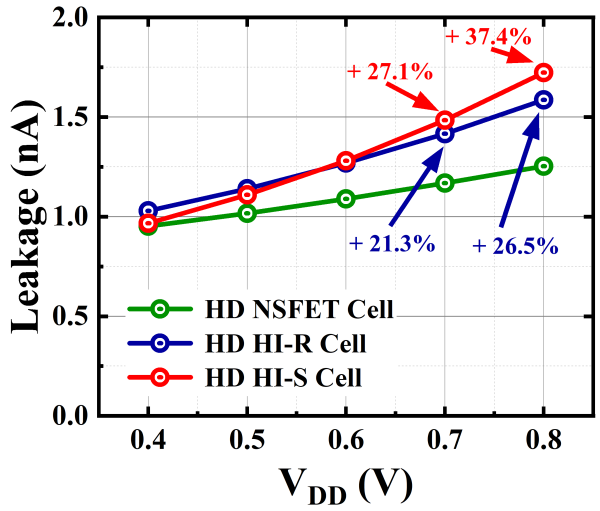


Fig. 9. Variation of leakage of HD 6T-SRAM cells with V_{DD}.

during write and read operations. On the other hand, the HD HI-R Cell exhibits a reduction in RT compared to the HD NSFET Cell due to the replacement of NSFETs by SL-FinFETs on the critical read path. Additionally, the use of p-type SL-FinFETs for the PU enhances the charging speed of the storage node during the writing operations, resulting in a reduced WT. The shortened RT and WT of HI-SRAM will enable them to operate at higher frequencies. However, it is important to note that the improvement in transient characteristic comes at the cost of higher power consumption, as shown in Fig. 9.

B. HP SRAM Optimization

Two distinct types of HP NSFET SRAM cells were designed by leveraging the flexibility of the NSFET width. The first type is the HP NSFET Cell for high-speed operations (HP-S NSFET Cell), where the AX and PD widths increase synchronously. The second type is HP NSFET Cell for robustness (HP-R NSFET Cell), where only the PD width is increased. The HP HI-SRAM cells achieve performance improvements by increasing the number of quantized Fin (see Fig. 6).

Fig. 10(a) shows the RT of the HP SRAM cells. With increasing AX and PD widths, the RT of the HP-S NSFET Cell decreases rapidly. When the cell height reaches 208 nm (with AX and PD widths of 21 nm), the RT reaches its minimum

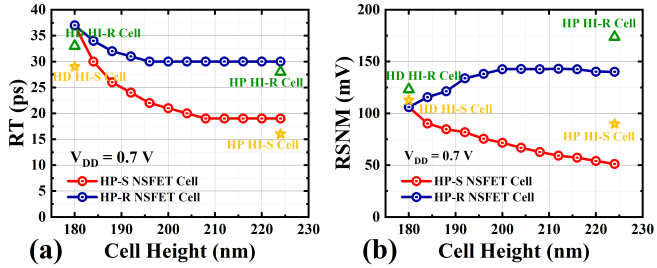


Fig. 10. Variation of (a) RT and (b) RSNM with cell height of HP 6T-SRAM cells at 0.7 V operating voltage.

value of 19 ps. However, further speed improvements are limited by the increased gate capacitance resulting from the continued increase in AX width. The charging time for the AX gate capacitance becomes the primary factor restricting further enhancements in speed. Regarding the HP-R NSFET Cell, the RT also slightly decreases due to the increased current on the critical read path. The HI-SRAM cells exhibit even better speed performance. The speed of the HD HI-S Cell is comparable to that of the HP-S NSFET Cell with a cell height of 186 nm. Moreover, the RT of the HP HI-S Cell continues to decrease by 15.8% below the limit value of the HP-S NSFET Cell. Although the speed of the HI-R Cell is inferior to that of the HP-S NSFET Cell, it still surpasses the speed of the HP-R NSFET Cell. The adoption of SL-FinFET, with its larger current brings a noteworthy breakthrough in speed compared to traditional NSFET SRAM cells.

Fig. 10(b) shows the RSNM of the HP SRAM cells. The RSNM of the HP-R NSFET Cell increases as the PD ratio grows due to the wider PD width. When the cell height reaches 200 nm (with a PD width of 17 nm), the HP-R NSFET Cell achieves a maximum RSNM of 143 mV. The HI-R Cells exhibit even better stability performance. The RSNM of the HD HI-R Cell is comparable to that of the HP-R NSFET Cell with a cell height of 188 nm. Furthermore, the RSNM of the HP HI-R Cell continues to increase by 21.6% beyond the limit value of the HP-R NSFET Cell. The hybrid integration of SL-FinFET and NSFET brings a larger transistor ratio and results in a notable breakthrough improvement in RSNM. On the other hand, the HP HI-S Cell and the HP-S NSFET Cell utilize larger AX widths but sacrifice the transistor ratio design, leading to a decrease in RSNM. The larger PU current of the HP HI-S Cell makes it more challenging to rewrite stored information, resulting in a smaller reduction in RSNM compared to the HP-S NSFET Cell.

C. SiGe Thickness Optimization in the Si/SiGe Epitaxy Process

The hybrid integration scheme enables the adjustment of the transistor ratio in the height dimension, overcoming the limitations of traditional method that relies on width design and resulting in a significant reduction in circuit area. This adjustment is achieved by modifying the SiGe thickness in the Si/SiGe epitaxy process. In SL-FinFET, the effective channel width (W_{eff}) is primarily determined by the height of the Fin, allowing changes in SiGe thickness to alter the W_{eff}. However, for NSFET, the W_{eff} is determined by the perimeter

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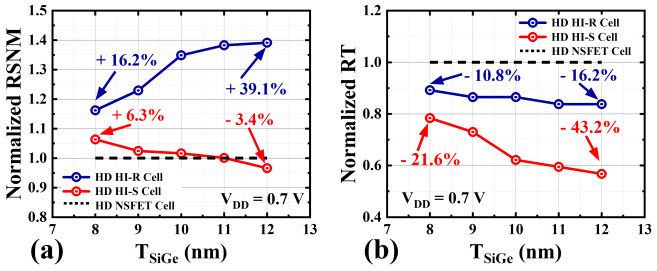


Fig. 11. (a) Normalized RSNM and (b) normalized RT versus T_{SiGe} of HD 6T-SRAM cells at 0.7 V operating voltage.

of Si nanosheets, so the variation in SiGe thickness does not affect the W_{eff} of NSFET. In other words, increasing the SiGe thickness during the Si/SiGe epitaxy process on one wafer may increase the ratio of SL-FinFET to NSFET without increasing the transistor occupied area, even though both kinds of the devices on the wafer have the same channel height. Fig. 11 shows the impact of SiGe thickness on the performance of HD SRAM cells. Since the SiGe thickness has little effect on the NSFET current, the performance of HD NSFET SRAM cells remains essentially unchanged. For the HD HI-R Cell, increasing the SiGe thickness from 8 to 12 nm leads to an improvement in RSNM from 16.2% to 39.1%. The PU ratio of the HD HI-R Cell continues to increase with SiGe thickness, further enhancing RSNM. Regarding the HD HI-S Cell, increasing the SiGe thickness from 8 to 12 nm results in a reduction in RT from 21.6% to 43.2%. The current of the SL-FinFET steadily increases with SiGe thickness, leading to further RT reduction in the HD HI-S Cell. Overall, increased SiGe thickness enhances the advantages of HI-SRAM cells.

VI. Conclusion

In summary, this article presents an innovative SRAM bitcell scheme that hybrid integrates both NSFET and SL-FinFET in one cell. This hybrid integration method is fully compatible with mainstream NSFET manufacturing processes, achieved through the precise formation of selective nanosheet channels. By harnessing the inherent transistor ratios provided by Si NSFET and SiGe SL-FinFET, this revolutionary scheme enables considerable performance enhancements while reducing the area. Compared to the HD NSFET SRAM cell, the HD HI-R Cell achieves a noteworthy 16.2% improvement in RSNM, while the HD HI-S Cell demonstrates notable reductions of 21.6% in RT and 28.6% in WT. The HP HI-S Cells even push the performance of SRAM cells to approach the theoretical limits set by regular NSFETs. Moreover, the hybrid integration scheme offers a promising pathway to optimize the transistor ratio in SRAM cells by adjusting the SiGe thickness in the transistors, thereby further enhancing their performance. The proposed technology provides one promising approach to scaling SRAM with higher performance and density for 3 nm and beyond nodes.

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